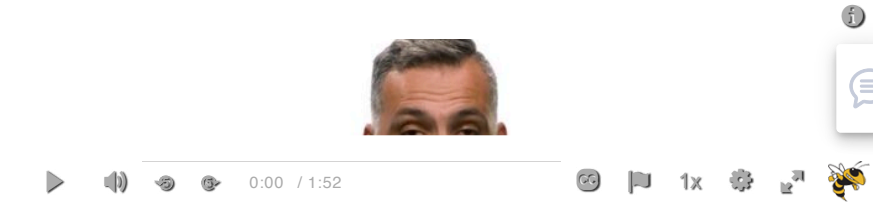


Lesson Eleven Overview



Throughout the lecture, we are giving you some “**food for thought**” -- these are questions we suggest you think about as you go through this lesson. You do not need to submit solutions.

Learning Objectives

Students will be able to:

- Understand the notion of network robustness in the context of random failures and attacks
- Learn about the problem of distributed search and how its efficiency depends on the network structure
- Model distributed synchronization phenomena on networks
- Become familiar with adaptive or co-evolutionary network phenomena

Required and Recommended Reading

Required Reading

- Chapter-8 (sections 8.1-8.3) from A-L. Barabási, [Network Science](http://www.networksciencebook.com/) (<http://www.networksciencebook.com/>), 2015.
- Chapter 20 - D. Easley and J. Kleinberg, [Networks, Crowds and Markets](https://www.amazon.com/Networks-Crowds-Markets-Reasoning-Connected/dp/0521195330) [↗](https://www.amazon.com/Networks-Crowds-Markets-Reasoning-Connected/dp/0521195330) (<https://www.amazon.com/Networks-Crowds-Markets-Reasoning-Connected/dp/0521195330>), Cambridge Univ Press, 2010 (also available [online](https://www.cs.cornell.edu/home/kleinber/networks-book)) [↗](https://www.cs.cornell.edu/home/kleinber/networks-book) (<https://www.cs.cornell.edu/home/kleinber/networks-book>)
- Chapter 21 (section 21.5) - D. Easley and J. Kleinberg, [Networks, Crowds and Markets](https://www.amazon.com/Networks-Crowds-Markets-Reasoning-Connected/dp/0521195330) [↗](https://www.amazon.com/Networks-Crowds-Markets-Reasoning-Connected/dp/0521195330) (<https://www.amazon.com/Networks-Crowds-Markets-Reasoning-Connected/dp/0521195330>), Cambridge Univ Press, 2010 (also available [online](https://www.cs.cornell.edu/home/kleinber/networks-book)) [↗](https://www.cs.cornell.edu/home/kleinber/networks-book) (<https://www.cs.cornell.edu/home/kleinber/networks-book>)

Recommended Reading

- [Synchronization in complex networks](https://www.sciencedirect.com/science/article/pii/S0370157308003384?casa_token=QxVjOU3HcM8AAAAA:RPNz2aIVi7cMQT3y0TtbHQDwM51cL7fGvK0ajWo7FLAIPH03468AqrhdXJDbOs2xxc_X5IFV4A) [↗](https://www.sciencedirect.com/science/article/pii/S0370157308003384?casa_token=QxVjOU3HcM8AAAAA:RPNz2aIVi7cMQT3y0TtbHQDwM51cL7fGvK0ajWo7FLAIPH03468AqrhdXJDbOs2xxc_X5IFV4A) (https://www.sciencedirect.com/science/article/pii/S0370157308003384?casa_token=QxVjOU3HcM8AAAAA:RPNz2aIVi7cMQT3y0TtbHQDwM51cL7fGvK0ajWo7FLAIPH03468AqrhdXJDbOs2xxc_X5IFV4A) by A.Arenas et al., 2008
- [Adaptive coevolutionary networks: a review](https://royalsocietypublishing.org/doi/abs/10.1098/rsif.2007.1229) [↗](https://royalsocietypublishing.org/doi/abs/10.1098/rsif.2007.1229) (<https://royalsocietypublishing.org/doi/abs/10.1098/rsif.2007.1229>) by T.Gross and B.Blasius, 2008

- [Co-evolutionary dynamics in social networks: A case study of twitter](https://link.springer.com/article/10.1186/s40649-015-0023-6)  (https://link.springer.com/article/10.1186/s40649-015-0023-6) by D.Antoniades and C.Dovrolis, 2015

Assignments

L11: Quiz

L11: Knowledge Check